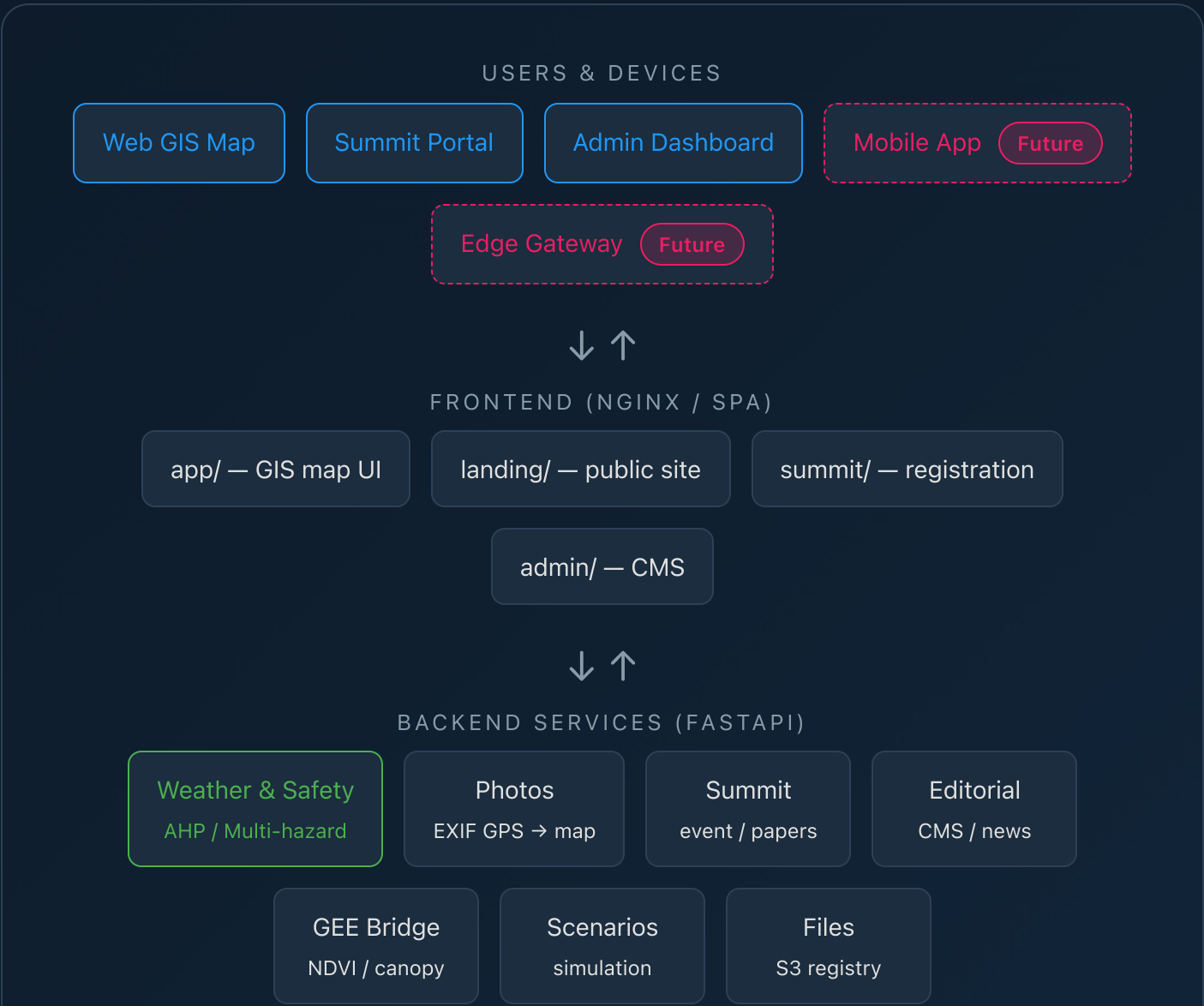


SylvaNexus (HiiForest) — Full Platform Architecture

A multi-service platform on AWS for forest science, safety early-warning, and community engagement in Pacific-Asia mountain forestry

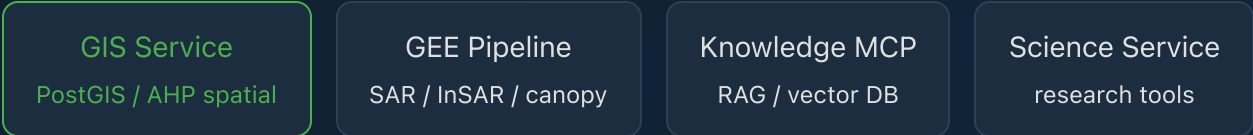
System Architecture Overview

All services run on a single EC2 instance (Tokyo ap-northeast-1) via Docker Compose, with S3 for object storage and GitHub Actions for CI/CD.

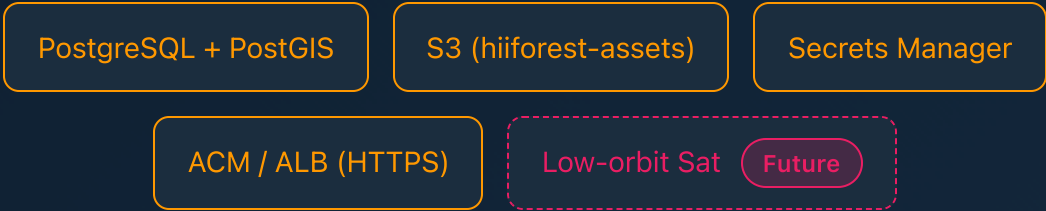




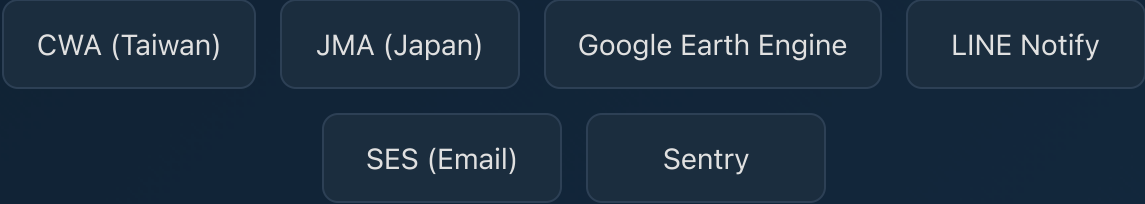
MICROSERVICES



DATA & INFRASTRUCTURE (AWS)



EXTERNAL DATA SOURCES



Platform Services



Forest Safety & AHP

Production

Multi-hazard early-warning using the AHP composite safety index (Rahmawati et al., 2025). Integrates real-time weather (CWA/JMA), terrain data (PostGIS), and 14 weighted safety criteria to produce spatial risk maps and crew advisories.



Forest Photo Pipeline

Production

Field photos with EXIF GPS metadata → S3 (original/medium/thumbnail) → map markers on the GIS frontend. Supports viewport-based queries and project filtering.

Endpoints: /landslide-risk, /safety-index-map,
/terrain-profile, /ahp-weights



Summit Registration

Production

Academic conference / event registration portal with paper upload, payment proof, and admin file management. Used for Taiwan-Japan forestry summits.



Editorial CMS

Production

Forest-science articles and news publishing system with image management via S3.



Google Earth Engine Bridge

Production

Satellite analysis: NDVI risk scoring, canopy height estimation, SAR-based landslide detection, InSAR deformation monitoring. Results stored in PostGIS and S3.



GIS Service (PostGIS)

Production

Spatial analysis microservice with PostGIS backend. Manages terrain_risk_grid, compartment boundaries, inventory data, and AHP safety index computation per grid cell.



Scenario Engine

Dev

Simulation engine for forest management scenarios — tree growth, harvest planning, carbon estimation. Pluggable engine architecture.



Knowledge MCP / RAG

Dev

Retrieval-augmented generation over forestry laws, research papers, and legal cases. Vector DB (pgvector) with document Q&A interface.



Unified File Registry

Production

Cross-module S3 file management API: presigned URLs, tagging, GPS metadata, RAG indexing status. Shared by photos, summit, editorial, and GIS modules.



Monitoring & Alerting

Production

GitHub Actions uptime monitor (every 5 min) + Sentry error tracking + LINE/Email safety advisories for field crews.

AWS Infrastructure



EC2 (ap-northeast-1)

t3.small instance running Docker Compose with 4 containers: backend, gis-service, frontend (nginx), postgres+PostGIS. Deployed via GitHub Actions → SSM Run Command.



S3 — hiiforest-assets

Object storage for field photos, GIS exports, summit papers/proofs, research PDFs, editorial images. Auto-tagged with module, project, GPS, and RAG status.



Secrets Manager

GEE service account keys, database credentials, and API keys stored securely. Mounted into containers at runtime.



ALB + ACM (HTTPS)

Application Load Balancer with AWS Certificate Manager TLS termination for hiiforest.com domain.



SES (Email)



GitHub Actions CI/CD

Amazon Simple Email Service for alert notifications and summit portal communications.

Automated deploy via SSM Run Command, browser E2E tests, uptime monitoring, and branch sync (research → main).

From Reference to Real-Time Field Action

The Gap We Are Closing

The AHP framework by Rahmawati et al. (2025) provides a critical reference for assessing natural-information risks to human life and property in forest operations. However, translating a static assessment framework into **real-time, actionable field support** requires bridging the gap between the GIS platform and the crew on the mountain.

Our goal is to make the GIS platform not just a map or a report, but a **living, interactive real-time assistance tool** that feeds safety intelligence back to the field — where it matters most.



Mobile App

Roadmap

A native mobile application for forest workers that provides:

- Real-time risk map with GPS positioning
- Push notifications for escalating hazards
- Check-in / check-out with location stamp
- Offline map tiles for no-signal areas
- Photo reporting with auto-GPS tagging



Low-Orbit Satellite Integration

Roadmap

For remote mountain areas without cellular coverage, low-orbit satellite (e.g., Starlink, Iridium) connectivity enables:

- Real-time weather data sync to field devices
- SOS / emergency signal from crew to platform
- Bi-directional safety status updates
- IoT sensor data uplink (rain gauge, wind mast, soil moisture)

- Crew safety status dashboard for supervisors

- Edge-cached risk grid refresh via satellite backhaul

Edge Computing Roadmap

Lightweight inference on field gateways for off-network resilience:

- Local cache of AHP risk grid for offline use
- On-device threshold alerts without cloud round-trip
- Federated sync when connectivity returns
- Sensor fusion (rain + wind + soil) at the edge

LLM-Powered Advisory Roadmap

AI-generated, locale-aware safety briefings:

- Gemini-powered crew briefing generation from AHP scores
- Multilingual (Chinese / Japanese / English) safety language
- Context-aware recommendations per work site
- Voice interface for hands-free field use

Shared Technology Stack — Related Projects

The same GIS + LLM infrastructure that powers SylvaNexus also supports other domain applications. These projects share the same AWS EC2 instance, PostGIS database, and nginx reverse proxy, demonstrating the extensibility of the platform architecture.

LegalShield-jp

A Japan-focused legal-aid GIS platform that aggregates public datasets (crime statistics, NPO/NGO directories, legal consultation offices) with user GPS location to instantly return "nearest

Forestry Knowledge MCP

A Model Context Protocol (MCP) server exposing a pgvector-backed corpus of forestry research papers and local documents to AI agents. Enables semantic search, citation network

support resources + local risk indicators" in one shot.

Built on the same FastAPI + PostGIS + pgvector stack, with LLM-powered document Q&A over Japanese legal texts. Shares EC2, PostgreSQL, and nginx with SylvaNexus.

Data sources: 警察庁 open data, e-Stat, 法テラス, 内閣府 NPO list, 国土数値情報

Repo: github.com/Fuilko/lawandbabysupport

analysis, and site-specific literature recommendations.

Uses bge-m3 embeddings with tsvector fallback. Designed to feed research context into the SylvaNexus safety scoring and advisory pipeline.

Tools: search_papers, search_local_docs, get_citation_network, recommend_for_site

Status: Skeleton — schema + MCP server done, awaiting OpenAlex harvest

Collaboration Invitation

To Researchers & Practitioners

The AHP forest-work-safety framework by Rahmawati et al. (2025) is the scientific foundation of our safety scoring engine. We are building the engineering layer that brings this framework from paper to the field — real-time, spatial, and actionable.

We welcome collaboration on:

- **Validation** of AHP criteria weights in Taiwan / Japan / Pacific-Asia forestry contexts
- **Expansion** of criteria for multi-hazard scenarios (landslide + heat + wind combined)
- **Field testing** of the mobile app and real-time alerting with forestry crews
- **Satellite / IoT integration** for remote-area connectivity
- **Academic publication** of the integrated GIS-AHP real-time platform

Platform: <https://hiiforest.com>

Full source: github.com/Fuilko/SaaSdocker

Public share: github.com/Fuilko/SylvaNexus-Forest-Safety

SylvaNexus / HiiForest — Forest Operations Safety Early-Warning Platform

Built for Pacific-Asia mountain forestry · 2026